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Guidelines

Treatment of Non-Metastatic Muscle-Invasive Bladder Cancer: AUA/ASCO/ASTRO/SUO Guideline

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SUMMARY

Purpose

Although representing approximately 25% of patients diagnosed with bladder cancer, muscle-invasive bladder cancer (MIBC) carries a significant risk of death that has not significantly changed in decades. Increasingly, clinicians and patients recognize the importance of multidisciplinary collaborative efforts that take into account survival and quality of life (QOL) concerns. For the first time for any type of malignancy, the American Urological Association (AUA), the American Society of Clinical Oncology (ASCO), the American Society for Radiation Oncology (ASTRO), and the Society of Urologic Oncology (SUO) collaborated on an evidence-based guideline. This guideline provides a risk-stratified clinical framework for the management of muscle-invasive urothelial bladder cancer and is designed to be used in conjunction with the associated treatment algorithm.

Methodology

The systematic review utilized to inform this guideline was conducted by a methodology team at the Pacific Northwest Evidence-based Practice Center. The original review was funded by the Agency for Healthcare Research and Quality (AHRQ), and a subsequent supplemental report was funded by the AUA to address additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in Ovid MEDLINE® (January 1990 to October 2014), the Cochrane Central Register of Controlled Trials (through September 2014), the Cochrane Database of Systematic Reviews (through September 2014), Health Technology Assessments (through Third Quarter 2014), the National Health Sciences Economic Evaluation Database (through Third Quarter 2014), and the Database of Abstracts of Reviews of Effects (through Third Quarter 2014) to capture published and gray literature. The methodology team searched for unpublished studies in clinical trial registries (ClinicalTrials.gov, Current Controlled Trials, ClinicalStudyResults.org and the World Health Organization International Clinical Trials Registry Platform) and regulatory documents (Drugs@FDA.gov and FDA Medical Devices Registration and Listing). A supplemental search of Ovid MEDLINE and Cochrane Central Register of Controlled Trials was conducted to capture additional published literature through February 2, 2016. The guideline underwent review in 2020. The updated search (July 1, 2016 to May 18, 2020) identified 2,005 abstracts, of which 38 met inclusion criteria. An additional review was performed in 2023. The updated search (May 2020 to November 2023) identified 3,739 abstracts, of which 46 met inclusion criteria. When sufficient evidence existed, the body of evidence for a particular treatment was assigned a strength rating of A (high), B (moderate) or C (low) for support of Strong, Moderate, or Conditional Recommendations. In the absence of sufficient evidence, additional information is provided as Clinical Principles and Expert Opinions.

GUIDELINE STATEMENTS

Initial Patient Evaluation and Counseling

1. Prior to treatment consideration, a full history and physical exam should be performed, including an exam under anesthesia, at the time of transurethral resection of bladder tumor (TURBT) for a suspected invasive cancer. (*Clinical Principle*)
2. Prior to muscle-invasive bladder cancer management, clinicians should perform a complete staging evaluation, including imaging of the chest and cross sectional imaging of the abdomen and pelvis with intravenous contrast if not contraindicated. Laboratory evaluation should include a comprehensive metabolic panel (complete blood count, liver function tests, alkaline phosphatase, and renal function). (*Clinical Principle*)
3. An experienced genitourinary pathologist should review the pathology of a patient when variant histology is suspected or if muscle invasion is equivocal (e.g., micropapillary, nested, plasmacytoid, neuroendocrine, sarcomatoid, extensive squamous or glandular differentiation). (*Clinical Principle*)
4. For patients with newly diagnosed muscle-invasive bladder cancer, curative treatment options should be discussed before determining a plan of therapy that is based on both patient comorbidity and tumor characteristics. Patient evaluation should be completed using a multidisciplinary approach. (*Clinical Principle*)
5. Prior to treatment, clinicians should counsel patients regarding complications and the implications of treatment on QOL (e.g., impact on continence, sexual function, fertility, bowel dysfunction, metabolic problems). (*Clinical Principle*)

Treatment

NEOADJUVANT/ADJUVANT CHEMOTHERAPY

6. Utilizing a multidisciplinary approach, clinicians should offer cisplatin-based neoadjuvant chemotherapy (NAC) to eligible radical cystectomy patients prior to cystectomy. (*Strong Recommendation; Evidence Level: Grade B*)
7. Clinicians should not prescribe carboplatin-based NAC for clinically resectable stage cT2-T4aN0 bladder cancer. Patients ineligible for cisplatin-based NAC should proceed to definitive locoregional therapy or clinical trial. (*Expert Opinion*)
8. Clinicians should perform radical cystectomy as soon as possible following a patient's completion of and recovery from NAC (ideally within 12 weeks unless medically inadvisable). (*Expert Opinion*)
9. Patients who have not received cisplatin-based NAC and have pT3-4and/or N+ disease at cystectomy should be offered adjuvant cisplatin-based chemotherapy or adjuvant immunotherapy. Patients who have received cisplatin-based chemotherapy and have pT2-4and/or N+ at cystectomy should be offered adjuvant immunotherapy. (*Moderate Recommendation; Evidence Level: Grade C*)

RADICAL CYSTECTOMY

10. Clinicians should offer radical cystectomy with bilateral pelvic lymphadenectomy for surgically eligible patients with resectable non-metastatic (M0) muscle-invasive bladder cancer. (*Strong Recommendation; Evidence Level: Grade B*)
11. When performing a standard radical cystectomy with curative intent, clinicians should remove the bladder, prostate, and seminal vesicles in males; clinicians should remove the bladder in females and should consider removal of adjacent reproductive organs based on individual disease characteristics and need to obtain negative margins. Organ sparing procedures in females should be considered based on disease location and characteristics on an individual basis. (*Clinical Principle*)
12. Clinicians should discuss and consider sexual function preserving procedures for patients with organ-confined disease and absence of bladder neck, urethra, and prostate (male) involvement. (*Moderate Recommendation; Evidence Level: Grade C*)

URINARY DIVERSION

13. In patients undergoing radical cystectomy, ileal conduit, continent cutaneous, and orthotopic neobladder urinary diversions should all be discussed. (*Clinical Principle*)
14. In patients receiving an orthotopic urinary diversion, clinicians must verify a negative urethral margin. (*Clinical Principle*)

PERIOPERATIVE SURGICAL MANAGEMENT

15. Clinicians should attempt to optimize patient performance status (PS) in the perioperative setting. (*Expert Opinion*)
16. Perioperative pharmacologic thromboembolic prophylaxis should be given to patients undergoing radical cystectomy. (*Strong Recommendation; Evidence Level: Grade B*)
17. In patients undergoing radical cystectomy μ -opioid antagonist therapy should be used to accelerate gastrointestinal recovery, unless contraindicated. (*Strong Recommendation; Evidence Level: Grade B*)
18. Patients should receive detailed teaching regarding care of urinary diversion prior to discharge from the hospital. (*Clinical Principle*)

PELVIC LYMPHADENECTOMY

19. Clinicians must perform a bilateral pelvic lymphadenectomy at the time of any surgery with curative intent. (*Strong Recommendation; Evidence Level: Grade B*)
20. When performing bilateral pelvic lymphadenectomy, clinicians should remove, at a minimum, the external and internal iliac and obturator lymph nodes (standard lymphadenectomy). (*Clinical Principle*)

Bladder Preserving Approaches

PATIENT SELECTION

21. For patients with newly diagnosed non-metastatic muscle-invasive bladder cancer who desire to retain their bladder, and for those with significant comorbidities for whom radical cystectomy is not a treatment option, clinicians should offer bladder preserving therapy when clinically appropriate. (*Clinical Principle*)

22. In patients under consideration for bladder preserving therapy, maximal debulking TURBT and assessment of multifocal disease/carcinoma in situ (CIS) should be performed. (*Strong Recommendation; Evidence Level: Grade C*)

MAXIMAL TURBT AND PARTIAL CYSTECTOMY

23. Patients with muscle-invasive bladder cancer who are medically fit and consent to radical cystectomy should not undergo partial cystectomy or maximal TURBT as primary curative therapy. (*Moderate Recommendation; Evidence Level: Grade C*)

PRIMARY RADIATION THERAPY

24. For patients with muscle-invasive bladder cancer, clinicians should not offer radiation therapy alone as a curative treatment. (*Strong Recommendation; Evidence Level: Grade C*)

MULTI-MODAL BLADDER PRESERVING THERAPY

25. For patients with muscle-invasive bladder cancer who have elected tri-modality therapy with organ preservation, clinicians should offer maximal TURBT followed by chemotherapy combined with external beam radiation therapy (EBRT). Planned cystoscopic surveillance per high-risk NMIBC schedule should be performed. (*Strong Recommendation; Evidence Level: Grade B*)
26. Radiation sensitizing chemotherapy should be included when using multimodal therapy with curative intent. (*Strong Recommendation; Evidence Level: Grade B*)
27. Following completion of bladder preserving therapy, clinicians should perform regular surveillance with computed tomography (CT) scans, cystoscopy, and urine cytology. (*Strong Recommendation; Evidence Level: Grade C*)

BLADDER PRESERVING TREATMENT FAILURE

28. In patients who are medically fit and have residual or recurrent muscle-invasive disease following bladder preserving therapy, clinicians should offer radical cystectomy with bilateral pelvic lymphadenectomy. (*Strong Recommendation; Evidence Level: Grade C*)
29. In patients who have a non-muscle invasive recurrence after bladder preserving therapy, clinicians may offer either local measures, such as TURBT with intravesical therapy, or radical cystectomy with bilateral pelvic lymphadenectomy. (*Moderate Recommendation; Evidence Level: Grade C*)

Patient Surveillance and Follow Up

IMAGING

30. Clinicians should obtain chest imaging and cross-sectional imaging of the abdomen and pelvis with CT or magnetic resonance imaging (MRI) at 6-12 month intervals for 2-3 years and then may continue annually. (*Expert Opinion*)

LABORATORY VALUES AND URINE MARKERS

31. Following therapy for muscle-invasive bladder cancer, patients should undergo laboratory assessment at three to six month intervals for two to three years and then annually thereafter. (*Expert Opinion*)
32. Following radical cystectomy in patients with a retained urethra, clinicians should monitor the urethral remnant for recurrence. (*Expert Opinion*)

PATIENT SURVIVORSHIP

33. Clinicians should discuss with patients how they are coping with their bladder cancer diagnosis and treatment and should recommend that patients consider participating in cancer support groups or consider receiving individual counseling. (*Expert Opinion*)
34. Clinicians should encourage bladder cancer patients to adopt healthy lifestyle habits, including smoking cessation, exercise, and a healthy diet, to improve long-term health and QOL. (*Expert Opinion*)

Variant Histology

35. In patients diagnosed with variant histology, clinicians should consider unique clinical characteristics that may require divergence from standard evaluation and management for urothelial carcinoma. (*Expert Opinion*)

INTRODUCTION

Methodology

SYSTEMATIC REVIEW

The systematic review utilized to inform this guideline was conducted by a methodology team at the Pacific Northwest Evidence-based Practice Center. The original review was funded by AHRQ,¹ and a subsequent supplemental report was funded by the AUA to address additional key questions and more recently published literature. A research librarian experienced in conducting literature searches for comparative effectiveness reviews searched in Ovid MEDLINE® (January 1990 to October 2014), the Cochrane Central Register of Controlled Trials (through September 2014), the Cochrane Database of Systematic Reviews (through September 2014), Health Technology Assessments (through Third Quarter 2014), the National Health Sciences Economic Evaluation Database (through Third Quarter 2014), and the Database of Abstracts of Reviews of Effects (through Third Quarter 2014) to capture published and gray literature. The methodology team searched for unpublished studies in clinical trial registries (ClinicalTrials.gov, Current Controlled Trials, ClinicalStudyResults.org and the World Health Organization International Clinical Trials Registry Platform) and regulatory documents (Drugs@FDA.gov and FDA Medical Devices Registration and Listing). Reference lists of relevant studies and previous systematic reviews were hand-searched for additional studies. Scientific information packets were solicited from drug and device manufacturers and via a notice published in the Federal Register. Initial Database searches resulted in 3,921 potentially relevant articles. After dual review of abstracts and titles, 295 articles were selected for full-text dual review, and 39 studies (in 41 publications) were determined to meet inclusion criteria and were included in this review. A supplemental search of Ovid MEDLINE and Cochrane Central Register of Controlled Trials was conducted to capture additional published literature through February 2, 2016.

In 2020, the MIBC guideline was updated through the AUA amendment process in which newly published literature is reviewed and integrated into previously published guidelines in an effort to maintain currency. The amendment allowed for the incorporation of additional literature released since the initial publication of this guideline in 2017. For this literature review the methodology team searched Ovid MEDLINE(R) ALL from July 1, 2016 to May 18, 2020 (overlapping with search dates for the 2016 review ending October 6, 2016), and eliminated duplicate abstracts reviewed for earlier reports. The literature search identified 2,005 abstracts, of which 38 met inclusion criteria. Two of these citations were from secondary publications of another study included in this update or a previous report. Seven abstracts reported RCTs, 29 observational studies, and 2 systematic reviews.

An additional update was performed in 2023. The updated search gathered literature from May 2020 to November 2023 using Ovid MEDLINE(R) ALL and eliminated duplicate abstracts reviewed for earlier reports. This review identified 3,739 abstracts, of which 46 met inclusion criteria. Eleven of these citations were from secondary publications of another study included in this update or a previous report. Nine studies were randomized controlled trials (RCTs), two non-randomized trials, 19 observational studies, and five systematic reviews.

DATA EXTRACTION AND DATA MANAGEMENT

The methodology team extracted the following information into evidence tables: study design; setting; inclusion and exclusion criteria; dose and duration of treatment for experimental and control groups; duration of follow up; number of subjects screened, eligible, and enrolled; population ES-5 characteristics (including age, race/ethnicity, sex, stage of disease, and functional status); results; adverse events; withdrawals due to adverse events; and sources of funding. Methodologists verified or calculated relative risks and associated 95% confidence intervals (CIs) based on the information provided (sample sizes and incidence of outcomes in each intervention group). Methodologists noted discrepancies between calculated and reported results when present. Data extraction for each study was completed by one investigator and independently reviewed for accuracy and completeness by a second investigator.

ASSESSMENT OF THE RISK OF BIAS OF INDIVIDUAL STUDIES

The methodology team assessed the risk of bias for RCTs and observational studies using criteria adapted from those developed by the U.S. Preventive Services Task Force.^{2, 3} These criteria were applied in conjunction with the approach recommended in the AHRQ Methods Guide³ for medical interventions. Two investigators independently assessed the risk of bias of each study. Discrepancies were resolved through discussion and consensus. Each study was rated as low, medium, or high risk of bias. Methodologists rated the quality of each RCT based on the methods used for randomization, allocation concealment, and blinding; the similarity of compared groups at baseline; whether attrition was adequately reported and acceptable; similarity in use of co-interventions; compliance with allocated treatments; the use of intent-to-treat analysis; and avoidance of selective outcomes reporting.² Methodologists rated the quality of each cohort study based on whether it enrolled a consecutive or random sample of patients meeting inclusion criteria; whether it evaluated comparable groups; whether rates of loss to follow up were reported and acceptable; whether it used accurate methods for ascertaining exposures, potential confounders, and outcomes; and whether it performed adjustment for important potential confounders (defined as a minimum of age, sex, tumor stage, and tumor grade).² Studies rated low risk of bias were considered to have no more than very minor methodological shortcomings with their results likely to be valid. Studies rated

medium risk of bias have some methodological shortcomings, but no flaw or combination of flaws judged likely to cause major bias. In some cases, the article did not report important information, making it difficult to assess its methods or potential limitations. The category of medium risk of bias is broad, and studies with this rating vary in their strengths and weaknesses; the results of some studies assessed to have medium risk of bias are likely to be valid, while others may be only possibly valid. Studies rated high risk of bias have significant flaws that may invalidate the results. They have a serious or fatal flaw or combination of flaws in design, analysis, or reporting; large amounts of missing information (including publication of only preliminary results in a subgroup of patients randomized); or serious discrepancies in reporting. Methodologists did not exclude studies rated as having high risk of bias a priori, but they were considered the least reliable when synthesizing the evidence, particularly when discrepancies between studies were present.

DETERMINATION OF EVIDENCE STRENGTH

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE)⁴ system was used to determine the aggregate evidence quality for each outcome, or group of related outcomes, informing Key Questions. GRADE defines a body of evidence in relation to how confident guideline developers can be that the estimate of effects as reported by that body of evidence is correct. Evidence is categorized as high, moderate, low and very low, and assessment is based on the aggregate risk of bias for the evidence base, plus limitations introduced as a consequence of inconsistency, indirectness, imprecision and publication bias across the studies. Upgrading of evidence is possible if the body of evidence indicates a large effect or if confounding factors would suggest either spurious effects or would reduce the demonstrated effect.

The AUA employs a 3-tiered strength of evidence system to underpin evidence-based guideline statements. Table 1 summarizes the GRADE categories, definitions and how these categories translate to the AUA strength of evidence categories. High certainty by GRADE translates to AUA A category strength of evidence, moderate to B, and both low and very low to C.

TABLE 1: Strength of Evidence Definition		
AUA Strength of Evidence Category	GRADE Certainty Rating	Definition
A	High	• Very confident that the true effect lies close to that of the estimate of the effect
B	Moderate	• Moderately confident in the effect estimate • The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
C	Low	• Confidence in the effect estimate is limited • The true effect may be substantially different from the estimate of the effect
	Very Low	• Very little confidence in the effect estimate • The true effect is likely to be substantially different from the estimate of effect

AUA NOMENCLATURE: LINKING STATEMENT TYPE TO EVIDENCE STRENGTH

The AUA nomenclature system explicitly links statement type to body of evidence strength, level of certainty, magnitude of benefit or risk/burdens, and the Panel’s judgment regarding the balance between benefits and risks/burdens (Table 2). **Strong Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is substantial. **Moderate Recommendations** are directive statements that an action should (benefits outweigh risks/burdens) or should not (risks/burdens outweigh benefits) be undertaken because net benefit or net harm is moderate. **Conditional Recommendations** are non-directive statements used when the evidence indicates that there is no apparent net benefit or harm or when the balance between benefits and risks/burden is unclear. All three statement types may be supported by any body of evidence strength grade. Body of evidence strength Grade A in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances and that future

research is *unlikely to change confidence*. Body of evidence strength Grade B in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *could change confidence*. Body of evidence strength Grade C in support of a Strong or Moderate Recommendation indicates that the statement can be applied to most patients in most circumstances but that better evidence *is likely to change confidence*. Body of evidence strength Grade C is only rarely used in support of a Strong Recommendation. Conditional Recommendations also can be supported by any evidence strength. When body of evidence strength is Grade A, the statement indicates that benefits and risks/burdens appear balanced, the best action depends on patient circumstances, and future research is *unlikely to change confidence*. When body of evidence strength Grade B is used, benefits and risks/burdens appear balanced, the best action also depends on individual patient circumstances and better evidence *could change confidence*. When body of evidence strength Grade C is used, there is uncertainty regarding the balance between benefits and risks/burdens, alternative strategies may be equally reasonable, and better evidence is *likely to change confidence*.

Where gaps in the evidence existed, the Panel provides guidance in the form of **Clinical Principles** or **Expert Opinion** with consensus achieved using a modified Delphi technique if differences of opinion emerged.⁵ A *Clinical Principle* is a statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature. *Expert Opinion* refers to a statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which there may or may not be evidence in the medical literature.

TABLE 2: AUA Nomenclature Linking Statement Type to Level of Certainty, Magnitude of Benefit or Risk/Burden, and Body of Evidence Strength			
	Evidence Strength A (High Certainty)	Evidence Strength B (Moderate Certainty)	Evidence Strength C (Low Certainty)
Strong Recommendation (Net benefit or harm substantial)	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is substantial Applies to most patients in most circumstances and future research is unlikely to change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is substantial Applies to most patients in most circumstances but better evidence could change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) appears substantial Applies to most patients in most circumstances but better evidence is likely to change confidence (rarely used to support a Strong Recommendation)
Moderate Recommendation (Net benefit or harm moderate)	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is moderate Applies to most patients in most circumstances and future research is unlikely to change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) is moderate Applies to most patients in most circumstances but better evidence could change confidence	Benefits > Risks/Burdens (or vice versa) Net benefit (or net harm) appears moderate Applies to most patients in most circumstances but better evidence is likely to change confidence

Conditional Recommendation (No apparent net benefit or harm)	Benefits = Risks/Burdens Best action depends on individual patient circumstances Future research unlikely to change confidence	Benefits = Risks/Burdens Best action appears to depend on individual patient circumstances Better evidence could change confidence	Balance between Benefits & Risks/Burdens unclear Alternative strategies may be equally reasonable Better evidence likely to change confidence
Clinical Principle	A statement about a component of clinical care that is widely agreed upon by urologists or other clinicians for which there may or may not be evidence in the medical literature		
Expert Opinion	A statement, achieved by consensus of the Panel, that is based on members' clinical training, experience, knowledge, and judgment for which may or may not be evidence in the medical literature		

PROCESS

The Muscle-Invasive Bladder Cancer Panel was created in 2014 by the American Urological Association Education and Research, Inc. (AUA). The Practice Guidelines Committee (PGC) of the AUA selected the Panel Chair who in turn appointed the Vice Chair. In a collaborative process, additional Panel members, including additional members of ASTRO, ASCO, and SUO, with specific expertise in this area were then nominated and approved by the PGC. The AUA conducted a thorough peer review process. The draft guideline document was distributed to 128 peer reviewers, 67 of which submitted comments. The Panel reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the guideline was submitted for approval to the PGC and Science and Quality Council. Then it was submitted to the AUA, ASTRO, ASCO, and SUO Board of Directors for final approval. Panel members received no remuneration for their work. This represents the first joint guidelines by these organizations.

The 2020 and 2024 amendments also underwent peer review. The draft 2020 amendment was distributed to 69 peer reviewers, 18 of whom submitted 38 comments. The 2024 amendment was distributed to 61 peer reviewers, 20 of whom submitted 67 comments. Following each amendment, Panels reviewed and discussed all submitted comments and revised the draft as needed. Once finalized, the amendments were submitted for approval.

Background

EPIDEMIOLOGY

There are 83,190 new cases of bladder cancer and 16,840 bladder cancer deaths estimated for 2024 in the U.S.⁶ Approximately 25% of newly diagnosed patients have muscle-invasive disease,^{7, 8} a rate that has not changed over the last 10 years based on data from the Surveillance, Epidemiology, and End Results (SEER) registry.⁹ In addition, up to 50% or more patients with high-risk non-muscle invasive bladder cancer (NMIBC) can progress to invasive disease. The male to female ratio is 3:1, and disease incidence increases with age. While rates of bladder cancer are higher in Caucasians than other ethnicities, disease specific survival is worse overall for African-Americans.^{8, 10}

ETIOLOGY

All of the factors that contribute to the development of bladder cancer are not completely understood, but exposure to carcinogens (e.g., tobacco smoke) is the primary cause with some impact from genetic susceptibility. Smoking tobacco is the most important and common risk factor and is estimated to contribute to the development of 50% of bladder tumors, with current smokers at higher risk than former smokers.^{11, 12} Former smoking increases the risk of bladder cancer by a factor of 2.2 (95% CI: 2.0 to 2.4), and current smoking by a factor of 4.1 (95% CI: 3.7 to 4.5) compared to never having smoked.¹¹ Second hand smoke can also increase the risk for the development of bladder cancer.¹³ Following smoking, another risk factor that predisposes to bladder cancer is occupational exposure to carcinogens, namely aromatic amines (benzidine, 4-aminobiphenyl, 2-naphthylamine, 4-chloro-o-toluidine), polycyclic aromatic hydrocarbons, and chlorinated hydrocarbons, which contribute to approximately 20% of all bladder cancers.¹⁴⁻¹⁶ Occupational exposure accounts for 25% of bladder cancer diagnoses in men and 11% in women.¹⁷

There are several other well-documented risk factors. Pelvic radiation for other malignancies increases the likelihood of developing bladder cancer with a hazard ratio (HR) of 1.7.¹⁸ In addition, exposure to *S. haematobium* infection is predominantly associated with an increased risk of squamous cell carcinoma of the bladder and is a significant clinical problem in many developing nations. This disease process is a much less common entity in the United States and not the major focus of this report.

Genetic predisposition to bladder cancer has been linked to genes involved in metabolism of carcinogens such as N-acetyl transferase and *GSTM1*-null genotypes.¹⁹ Large genome-wide association studies have also found sequence variants that can increase the risk for bladder cancer, such as subjects with urea transporter gene *SLC14A* that is associated with renal urine concentration, and thus with variations in contact of carcinogens with urothelial surfaces.²⁰⁻²⁴

Urothelial carcinoma is often multifocal with a high rate of recurrence; the exact etiology of this characteristic is currently unknown. Two of the most commonly held theories: 1) a genetic field defect exists with multiple new tumors spontaneously arising or, 2) the local reimplantation of tumor cells occurs with tumor resection. Evidence suggests that tumor reimplantation or submucosal migration may be early mechanisms for multifocality.²⁵ Multifocal tumors as well as upper tract and lower tract lesions arising in one individual may demonstrate clonality.²⁶

PROGNOSIS

The overall prognosis of patients with MIBC has not changed in the last 30 years. In patients who undergo cystectomy, systemic recurrence rates vary by stage, but range from 20-30% for pathologic stage pT2, 40% for pT3, >50% for pT4 and approximately 70% for node-positive disease.^{27, 28} Most recurrences will occur within the first two to three years after cystectomy, and at this time, most patients with recurrence after cystectomy are not cured with current systemic therapies.²⁹

A pooled analysis of multiple prospective Radiation Therapy Oncology Group (RTOG) protocols evaluating bladder preserving combined-modality therapy for MIBC with a median follow up of 4.3 years found the 5- and 10-year overall survival rates were 57% and 36%, respectively, and the 5- and 10-year disease specific survival rates were 71% and 65%, respectively.³⁰

The dominant pathologic predictors for recurrence and survival are tumor stage and nodal status. Other prognostic factors include gender, presence of hydronephrosis, lymphovascular invasion, soft tissue margin status, and molecular subtyping characteristics.³¹⁻³⁶ Variant histology has become better described and recognized, and the treatment for these cancers may vary from conventional urothelial carcinoma. There is also a significant impact of treatment choices on outcome with the type and timing of therapy playing an important role.^{37, 38}

SCOPE

This evidence-based guideline for clinically non-metastatic muscle-invasive urothelial bladder cancer (cT2-T4N0M0) focuses on the evaluation, treatment, and surveillance of MIBC and is guided toward curative intent. The treatment of patients with clinically evident metastatic bladder cancer is outside the context of this guideline and will not be discussed. Optimal initial evaluation of patients with MIBC, including imaging and proper staging, are discussed. The role of radical cystectomy and bilateral pelvic lymphadenectomy is defined. Bladder preserving regimens such as a multi-modal approach that combines maximal TURBT, chemotherapy and radiation therapy as well as partial cystectomy, radiation alone and maximal TURBT alone, are assessed.

In addition, this guideline will discuss QOL aspects of care and the importance of careful patient counseling. The guidelines will also address timing and mode of testing used in surveillance of disease. Finally, there will be a section devoted to variant histology and the current unique aspects of care for certain non-urothelial cancers of the bladder.

GUIDELINE STATEMENTS

Initial Patient Evaluation & Counseling

GUIDELINE STATEMENT 1

Prior to treatment consideration, a full history and physical exam should be performed, including an exam under anesthesia, at the time of transurethral resection of bladder tumor (TURBT) for a suspected invasive cancer. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 2

Prior to muscle-invasive bladder cancer management, clinicians should perform a complete staging evaluation, including imaging of the chest and cross sectional imaging of the abdomen and pelvis with intravenous contrast if not contraindicated. Laboratory evaluation should include a comprehensive metabolic panel (complete blood count, liver function tests, alkaline phosphatase, and renal function). (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 3

An experienced genitourinary pathologist should review the pathology of a patient when variant histology is suspected or if muscle invasion is equivocal (e.g., micropapillary, nested, plasmacytoid, neuroendocrine, sarcomatoid, extensive squamous or glandular differentiation). (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 4

For patients with newly diagnosed muscle-invasive bladder cancer, curative treatment options should be discussed before determining a plan of therapy that is based on both patient comorbidity and tumor characteristics. Patient evaluation should be completed using a multidisciplinary approach. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 5

Prior to treatment, clinicians should counsel patients regarding complications and the implications of treatment on QOL (e.g., impact on continence, sexual function, fertility, bowel dysfunction, metabolic problems). (*Clinical Principle*)

Discussion

Treatment

NEOADJUVANT/ADJUVANT CHEMOTHERAPY

GUIDELINE STATEMENT 6

Utilizing a multidisciplinary approach, clinicians should offer cisplatin-based neoadjuvant chemotherapy (NAC) to eligible radical cystectomy patients prior to cystectomy. (*Strong Recommendation; Evidence Level: Grade B*)

Discussion

GUIDELINE STATEMENT 7

Clinicians should not prescribe carboplatin-based NAC for clinically resectable stage cT2-T4aN0 bladder cancer. Patients ineligible for cisplatin-based NAC should proceed to definitive locoregional therapy or clinical trial. (*Expert Opinion*)

Discussion

GUIDELINE STATEMENT 8

Clinicians should perform radical cystectomy as soon as possible following a patient’s completion of and recovery from NAC (ideally within 12 weeks unless medically inadvisable). (*Expert Opinion*)

Discussion

GUIDELINE STATEMENT 9

Patients who have not received cisplatin-based NAC and have pT3-4and/or N+ disease at cystectomy should be offered adjuvant cisplatin-based chemotherapy or adjuvant immunotherapy. Patients who have received cisplatin-based chemotherapy and have pT2-4and/or N+ at cystectomy should be offered adjuvant immunotherapy. (*Moderate Recommendation; Evidence Level: Grade C*)

Discussion

RADICAL CYSTECTOMY

GUIDELINE STATEMENT 10

Clinicians should offer radical cystectomy with bilateral pelvic lymphadenectomy for surgically eligible patients with resectable non-metastatic (M0) muscle-invasive bladder cancer. (*Strong Recommendation; Evidence Level: Grade B*)

Discussion

GUIDELINE STATEMENT 11

When performing a standard radical cystectomy with curative intent, clinicians should remove the bladder, prostate, and seminal vesicles in males; clinicians should remove the bladder in females and should consider removal of adjacent reproductive organs based on individual disease characteristics and need to obtain negative margins. Organ sparing procedures in females should be considered based on disease location and characteristics on an individual basis. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 12

Clinicians should discuss and consider sexual function preserving procedures for patients with organ-confined disease and absence of bladder neck, urethra, and prostate (male) involvement. (*Moderate Recommendation; Evidence Level: Grade C*)

Discussion

URINARY DIVERSION

GUIDELINE STATEMENT 13

In patients undergoing radical cystectomy, ileal conduit, continent cutaneous, and orthotopic neobladder urinary diversions should all be discussed. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 14

In patients receiving an orthotopic urinary diversion, clinicians must verify a negative urethral margin. (*Clinical Principle*)

Discussion

PERIOPERATIVE SURGICAL MANAGEMENT

GUIDELINE STATEMENT 15

Clinicians should attempt to optimize patient performance status (PS) in the perioperative setting. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 16

Perioperative pharmacologic thromboembolic prophylaxis should be given to patients undergoing radical cystectomy. *(Strong Recommendation; Evidence Level: Grade B)*

Discussion

GUIDELINE STATEMENT 17

In patients undergoing radical cystectomy μ -opioid antagonist therapy should be used to accelerate gastrointestinal recovery, unless contraindicated. *(Strong Recommendation; Evidence Level: Grade B)*

Discussion

GUIDELINE STATEMENT 18

Patients should receive detailed teaching regarding care of urinary diversion prior to discharge from the hospital. *(Clinical Principle)*

Discussion

PELVIC LYMPHADENECTOMY

GUIDELINE STATEMENT 19

Clinicians must perform a bilateral pelvic lymphadenectomy at the time of any surgery with curative intent. *(Strong Recommendation; Evidence Level: Grade B)*

Discussion

GUIDELINE STATEMENT 20

When performing bilateral pelvic lymphadenectomy, clinicians should remove, at a minimum, the external and internal iliac and obturator lymph nodes (standard lymphadenectomy). *(Clinical Principle)*

Discussion

Bladder Preserving Approaches

PATIENT SELECTION

A multi-modal bladder preserving approach with its merits and disadvantages should be discussed in each individual case. The studies that support bladder preserving strategies generally have highly select patient populations. There are currently no randomized trials comparing NAC and radical cystectomy versus multi-modality bladder preserving therapies. In reviewing the available studies regarding multi-modal bladder preserving protocols that employ TURBT, radiation therapy, and chemotherapy for carefully selected patients, the Panel found no strong evidence to determine whether or not immediate cystectomy improved survival when compared to initial bladder sparing protocols that employ salvage cystectomy as therapy for persistent bladder cancer.^{126-132, 134} In addition, no high quality evidence directly compares QOL between the different treatment options; instead a number of studies report on health-related QOL outcomes and draw

comparisons to other therapies. The Panel also recognizes that other non-multi-modal bladder–preserving regimens, although having less oncologic efficacy as well as less data, do exist and may be a reasonable option for certain patients, especially those who have poorer PS.

GUIDELINE STATEMENT 21

For patients with newly diagnosed non-metastatic muscle-invasive bladder cancer who desire to retain their bladder, and for those with significant comorbidities for whom radical cystectomy is not a treatment option, clinicians should offer bladder preserving therapy when clinically appropriate. (*Clinical Principle*)

Discussion

GUIDELINE STATEMENT 22

In patients under consideration for bladder preserving therapy, maximal debulking TURBT and assessment of multifocal disease/carcinoma in situ (CIS) should be performed. (*Strong Recommendation; Evidence Level: Grade C*)

Discussion

MAXIMAL TURBT AND PARTIAL CYSTECTOMY

GUIDELINE STATEMENT 23

Patients with muscle-invasive bladder cancer who are medically fit and consent to radical cystectomy should not undergo partial cystectomy or maximal TURBT as primary curative therapy. (*Moderate Recommendation; Evidence Level: Grade C*)

Discussion

PRIMARY RADIATION THERAPY

GUIDELINE STATEMENT 24

For patients with muscle-invasive bladder cancer, clinicians should not offer radiation therapy alone as a curative treatment. (*Strong Recommendation; Evidence Level: Grade C*)

Discussion

MULTI-MODAL BLADDER PRESERVING THERAPY

GUIDELINE STATEMENT 25

For patients with muscle-invasive bladder cancer who have elected tri-modality therapy with organ preservation, clinicians should offer maximal TURBT followed by chemotherapy combined with external beam radiation therapy (EBRT). Planned cystoscopic surveillance per high-risk NMIBC schedule should be performed. (*Strong Recommendation; Evidence Level: Grade B*)

Discussion

GUIDELINE STATEMENT 26

Radiation sensitizing chemotherapy should be included when using multimodal therapy with curative intent. (*Strong Recommendation; Evidence Level: Grade B*)

Discussion

GUIDELINE STATEMENT 27

Following completion of bladder preserving therapy, clinicians should perform regular surveillance with computed tomography (CT) scans, cystoscopy, and urine cytology. **(*Strong Recommendation; Evidence Level: Grade C*)**

Discussion

BLADDER PRESERVING TREATMENT FAILURE

GUIDELINE STATEMENT 28

In patients who are medically fit and have residual or recurrent muscle-invasive disease following bladder preserving therapy, clinicians should offer radical cystectomy with bilateral pelvic lymphadenectomy. **(*Strong Recommendation; Evidence Level: Grade C*)**

Discussion

GUIDELINE STATEMENT 29

In patients who have a non-muscle invasive recurrence after bladder preserving therapy, clinicians may offer either local measures, such as TURBT with intravesical therapy, or radical cystectomy with bilateral pelvic lymphadenectomy. **(*Moderate Recommendation; Evidence Level: Grade C*)**

Discussion

Patient Surveillance & Follow Up

IMAGING

GUIDELINE STATEMENT 30

Clinicians should obtain chest imaging and cross sectional imaging of the abdomen and pelvis with CT or magnetic resonance imaging (MRI) at 6-12 month intervals for 2-3 years and then may continue annually. **(*Expert Opinion*)**

Discussion

LABORATORY VALUES AND URINE MARKERS

GUIDELINE STATEMENT 31

Following therapy for muscle-invasive bladder cancer, patients should undergo laboratory assessment at three to six month intervals for two to three years and then annually thereafter. **(*Expert Opinion*)**

Discussion

GUIDELINE STATEMENT 32

Following radical cystectomy in patients with a retained urethra, clinicians should monitor the urethral remnant for recurrence. **(*Expert Opinion*)**

Discussion

PATIENT SURVIVORSHIP

GUIDELINE STATEMENT 33

Clinicians should discuss with patients how they are coping with their bladder cancer diagnosis and treatment and should recommend that patients consider participating in a cancer support group or consider receiving individual counseling. *(Expert Opinion)*

Discussion

GUIDELINE STATEMENT 34

Clinicians should encourage bladder cancer patients to adopt healthy lifestyle habits, including smoking cessation, exercise, and a healthy diet, to improve long-term health and QOL. *(Expert Opinion)*

Discussion

Variant Histology

GUIDELINE STATEMENT 35

In patients diagnosed with variant histology, clinicians should consider unique clinical characteristics that may require divergence from standard evaluation and management for urothelial carcinoma. *(Expert Opinion)*

Discussion

FUTURE DIRECTIONS

Several key areas of future research need emphasis to improve clinical care and provide a path to better patient outcomes with invasive bladder cancer.

Detection and markers. Improved imaging modalities to better locally stage tumors and define extent of disease are needed. This includes cystoscopic and radiographic imaging of local disease and more effective and accurate evaluation techniques of regional lymphatics and distant sites. The role of MRI, vesical imaging-reporting and data system (VI-RADS) for local staging and defining the role of PET imaging, the best PET imaging agent, and the investigation/validation of other novel technologies are deemed high-priority.

Urine cytology can be used to monitor for recurrence after TURBT and cystectomy, but difficulties with interpretation after urinary diversion have limited its usefulness after bladder removal. Radiation therapy can alter the appearance of shed cells and oftentimes result in atypical results. Current urinary markers have a limited role in the routine monitoring for recurrence of urothelial carcinoma after radical cystectomy due to false positive rate. Future studies should focus on the development of urinary and serum-based markers that can be used to identify early urothelial based and/or distant recurrences.

Increased knowledge gained from comprehensive genetic studies of invasive bladder cancer should be utilized to identify and validate markers that could be used to guide diagnosis and therapeutic decision making. This would include the identification of prognostic markers capable of stratifying patients at risk for advanced disease, predictive markers for the response to chemotherapeutic/immunotherapeutic agents as well as radiation-based therapies. In addition, further studies are needed to evaluate and validate the prognostic and predictive information obtained from novel molecular classifications of bladder cancer.

Therapy. The rapid introduction of novel immunotherapeutic agents for treatment of bladder cancer has begun to show promise. Phase II and III studies have now demonstrated significant antitumor activity of the anti-PD-1 and anti-PDL-1 antibodies in the metastatic setting. Additional studies are needed to further define the role of these agents alone or in combination with other therapies for all stages of bladder cancer.

In addition, further studies are needed to better integrate multi-modal therapy in patients with invasive bladder cancer. Specific examples include the role for AC or immunotherapy in patients who have previously received NAC followed by surgery but still possess high-risk pathology (residual invasive disease or regional lymph node involvement) and the role of immunotherapy in bladder preservation. A phase I trial investigating the addition of concurrent immune-checkpoint inhibition to chemoradiation for bladder preservation in patients with muscle-invasive disease found high rates of metastasis-free and overall survival, and a phase III trial (NCT03775265) investigating the role of atezolizumab with

chemoradiation recently completed enrollment and will provide further data regarding the role of immunotherapy with bladder preservation.²⁹¹ Additionally, the role of radiation in patients undergoing radical cystectomy for T3 and T4 disease, including the use of intraoperative radiation therapy, is yet to be clearly defined.

Robotic cystectomy has been adopted as a surgical option for the treatment of patients with invasive bladder cancer with the hope that it will improve the morbidity associated with radical cystectomy. RCTs have shown decreased blood loss with robotic cystectomy as compared to open cystectomy with no difference in complications, length of hospitalization, readmission rates, pain, QOL, or postoperative mortality and no difference in short-term progression-free survival.^{292, 293} Long-term data is needed to demonstrate the oncologic efficacy, potential for improved clinical outcomes, and QOL using this technology compared to standard open techniques.

Tissue regenerative technology continues to advance, stimulating the hope that organ replacement may be available in the future. Support of basic and translational research is needed to move tissue regeneration forward into clinical use for patients who require bladder removal for invasive bladder cancer.

The currently unpublished SWOG S1011 trial compared extended LND with standard LND and found no significant difference in DFS or overall survival.²¹⁹ However, presentation of unpublished long-term follow-up of the German LEA AUO AB 25/02 trial suggests improvement in survival associated with an extended LND.²²⁰

In addition, studies emphasizing PROs after treatment for invasive bladder cancer are needed. This information is necessary to help further support patient centered outcomes and identify specific areas of treatment that require further attention to improve patient QOL.

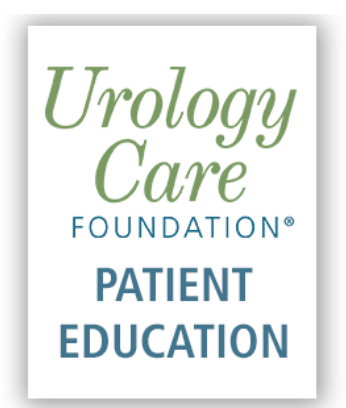
Surveillance. Finally, the optimal strategies for surveillance after definitive treatment for invasive bladder cancer to identify pelvic, distant, and urothelial recurrences need to be defined. The role of specific imaging tests and laboratory studies as well as their appropriate interval has yet to be established. Some evidence suggests a potential role for circulating tumor cell DNA (ctDNA) in detecting recurrence and progression following radical cystectomy. Future studies are needed to further define the potential role of ctDNA for surveillance.

ABBREVIATIONS

5-FU	5-fluorouracil
AC	Adjuvant chemotherapy
AHRQ	Agency for Healthcare Research and Quality
ASCO	American Society of Clinical Oncology
ASTRO	American Society for Radiation Oncology
AUA	American Urological Association
CIS	Carcinoma in situ
ctDNA	Circulating tumor cell DNA
CI	Confidence interval
CMV	Cisplatin, methotrexate and vinblastine
CT	Computed tomography
ddMVAC	Dose-dense methotrexate, vinblastine, Adriamycin, and cisplatin
DFS	Disease free survival
EBRT	External beam radiation therapy
GC	Gemcitabine and cisplatin

GRADE	Grading of Recommendations Assessment, Development, and Evaluation
HR	Hazard ratio
LND	Lymph node dissection
MIBC	Muscle invasive bladder cancer
MRI	Magnetic resonance imaging
MVAC	Methotrexate, vinblastine, Adriamycin, and cisplatin
NAC	Neoadjuvant chemotherapy
NMIBC	Non-muscle invasive bladder cancer
PGC	Practice Guidelines Committee
PS	Performance status
RCT	Randomized controlled trial
RTOG	Radiation Therapy Oncology Group
SEER	Surveillance, Epidemiology, and End Results
SUO	Society of Urologic Oncology
TURBT	Transurethral resection of bladder tumor
QOL	Quality of life
VI-RADS	Vesical imaging-reporting and data system

Tools and Resources



[Bladder Cancer Filtered Resources](#)

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